Palm OS Developer Certification

Sample Questions for Developer Certification

For a version of the sample questions without the answers listed, see Sample Questions.

1. Which statement is consistent with the Palm OS application design philosophy?

   A. More handheld RAM ensures more powerful applications.
   B. Powerful applications tend to have more complete feature sets.
   C. Mimicking familiar desktop user interface elements makes it easier for users to learn a new application.
   D. A simple feature set should be employed to be consistent with the way users use handhelds.

   ### Answer

   D. The handheld is not a small laptop PC; it is intrinsically something else. Handheld application design should focus on speed and ease of use, limiting features to only what is necessary.

   ### References

   For more information on application design, see the books *Zen of Palm: Designing Products for Palm OS*® and Palm OS® User Interface Guidelines, the training courses Introduction to Palm OS Application Development (online) and Palm OS Programming Essentials (instructor-led), and Article 1726 from the Developer Knowledge Base.

2. A Palm OS application requires sharing global data with a shared library. Which three mechanisms should be used? (Choose three.)

   A. Notifications
   B. Launch codes
   C. Global variables
   D. Feature pointers
   E. Application preferences

   ### Answer

   A, **C**, and **D**. Launch codes can only be sent to applications, not to shared libraries; notifications can be sent to any code resource, including shared libraries.

   Because an application's global data is not generally available while the application is processing a launch code, a feature pointer is the easiest way for an application make global variables available to shared libraries.

   ### References

   For more information on using shared libraries, see *Palm OS Programmer's Companion* and

the developer training class Advanced Palm OS Development and Debugging.

3. A Palm OS Solution has been developed and relies on a unique identifier for each handheld on which it is used. The solution must be compatible with all Palm Powered™ handhelds. Which two statements are true? (Choose two.)

   A. The HotSync user name is unique for all users and can be used as the unique identifier
   B. A unique handheld ID can be obtained at www.palmos.com and programmed into the handheld.
   C. The Palm IIIe is an example of a handheld that has no unique identifier of serial number accessible from software.
   D. There is no guarantee that a Palm Powered™ handheld has a unique identifier or serial number accessible from software
   E. All Palm Powered™ handhelds have a unique serial number programmed into the handheld and can be accessed by software.

   Answer

   C and **D**. Although Palm OS provides the function `SysGetROMToken`, you should always check the value returned, because handhelds manufactured earlier than the Palm III handheld do not have a ROM serial number.

   References

   For more information on Palm system support, see *Palm OS Programmer's Companion*. For information on Palm Powered handhelds, see the Palm OS Platform Hardware Development web page. See also the training course Palm OS Programming Essentials. See also Knowledge Base articles 2389, 1099.

4. What is the correct sequence for these form-related functions?

   A. `FrmDrawForm, FrmInitForm, FrmEraseForm, FrmDeleteForm`
   B. `FrmDrawForm, FrmInitForm, FrmDeleteForm, FrmEraseForm`
   C. `FrmInitForm, FrmDrawForm, FrmEraseForm, FrmDeleteForm`
   D. `FrmInitForm, FrmDrawForm, FrmDeleteForm, FrmEraseForm`

   Answer

   C. `FrmInitForm` loads and initializes a form resource, but it does not draw the form. `FrmDrawForm` draws the form and all the objects in the form. `FrmEraseForm` removes the form from the handheld display, but does not release the memory occupied by the form. `FrmDeleteForm` releases the memory used by the form and the form objects.

   References

   For more information on the Form API functions, see *Palm OS Programmer's API Reference* and the training class Palm OS Programming Essentials. See also Knowledge Base article 1209.

5. Which two statements are true about records? (Choose two.)

A. `DmRemoveRecord` disposes of a record's data chunk.
B. `DmArchiveRecord` sets the archive bit in the record's header.
C. `DmDeleteRecord` removes the record's header from the database.
D. `DmDeleteRecord` should not be used in an application with out a conduit.
E. `DmDetachRecord` causes a record to be deleted during the next HotSync operation.

Answer

A and **D**. The `DmArchiveRecord` function actually sets the delete bit to indicate that the record should be saved on the PC before being removed from the handheld. The `DmDeleteRecord` function deletes the record's chunk, but leaves the record entry in the header. The `DmDetachRecord` function detaches the record from the database, but does not delete the record's chunk.

References

For more information on the Data Manager API functions, see *Palm OS Programmer's API Reference* and the training class Palm OS Programming Essentials. See the Memo Pad sample application for an example of these functions in use.

6. Which three types of information does the HotSync Manager retrieve from the `GetConduitInfo` entry point? (Choose three.)

A. The conduit's default action.
B. The conduit's version number.
C. File link details from the conduit.
D. The display name of the conduit.
E. Whether MFC was used to build the conduit, and if so, which version.

Answer

A, **D**, and **E**. `GetConduitInfo` returns the values specified by the `ConduitInfoEnum` constants: conduit name, MFC version, and default action.

References

For more information on `GetConduitInfo`, see *C/C++ Sync Suite Reference*, *COM Sync Suite Reference* and the training class Creating Palm OS Conduits on Windows in C/C++

7. Your conduit must display a dialog box when a user selects your conduit and clicks **Change** in the HotSync Manager's Custom dialog box. The dialog must also display the user name in it.

Which entry point in your conduit is called to display the dialog box?

A. `CfgConduit`
B. `GetConduitName`
C. `GetConduitInfo`
D. `ConfigureConduit`
E. `ConfigureSubscription`

Answer

A. `CfgConduit` is an optional entry point that presents a dialog allowing the user to configure the conduit.

References

For more information on conduit entry points, see *C/C++ Sync Suite Companion*, *COM Sync Suite Companion* and the training class Creating Palm OS Conduits on Windows in C/C++

8. Given the MainFooButton is an ID for a valid button with the usable bit set to false:

```
#define MainForm      1000
#define MainFooButton 1001
```

And given this code:

```
frmP = FrmGetActiveForm();
FrmHideObject (frmP, MainFooButton);
```

Emulator has OS 4.0 Debug ROM and all debug checks on. What will this code do?

A. Generate an error at compile time
B. Crash on a handheld; crash on Palm OS Emulator
C. Do nothing on a handheld; crash on Palm OS Emulator
D. Hide the button on a handheld; crash on Palm OS Emulator

Answer

B. `FrmHideObject` requires the second argument to be an *index* of an object in the form.

References

For more information on the Form API functions, see *Palm OS Programmer's API Reference* and the training class Palm OS Programming Essentials. For information on using Palm OS Emulator, see *Using Palm OS Emulator*.

9. Given the following data structure in a Palm OS application:

```
typedef struct myCustomDataType {
   UInt8  one;
   UInt8  two;
   UInt16 three;
   UInt32 four;
   Char   fixedArray [20];
}myCustomeDataType;
```

A record of this structure is transferred using a Windows conduit. Which structure members need to have their endianness corrected?

A. All of the members
B. `three` and `four`
C. Only `fixedArray`
D. Only the integer types
E. None of the members
F. `three`, `four`, and `fixedArray`

Answer

B. Only multi-byte integer data items need to have endianness corrected.

References

For more information on transferring data between the handheld and Windows, see *C/C++ Sync Suite Reference* and the training class Creating Palm OS Conduits on Windows in C/C++

10. Given the following requirements for a conduit:

- The corresponding Palm application must handle the SysAppLaunchCmdSyncNotify launch code to perform post-sync operations
- Data modifications will be made on both the handheld and the desktop, though testing has indicated that a large majority of users prefer to enter data on the handheld.
- After the data has been synchronized, the data may be posted to a corporate accounting system on the desktop if the user has selected the Post Data menu option on the handheld.

Which is the ideal synchronization method and why?

A. Transaction-based conduit, because the conduit will not slow down the HotSync operation.
B. Mirror image conduit, because data conflicts cannot occur in this scenario.
C. Backup conduit, because the desktop application can parse the .pdb file stored on the handheld.
D. Mirror image conduit, because data manipulation may occur on either the handheld or the desktop.
E. Transaction-based conduit, because the user may specify the Post Data option at any time.

Answer

D. A mirror image conduit is ideal for applications that run on both the desktop and the handheld. A conduit of this type allows modifications on both and makes the data identical on both.

References

For information on synchronization types, see *Introduction to Conduit Development* and the training class Creating Palm OS Conduits on Windows in C/C++. See also Knowledge Base article 1679.

11. Given:

```
UInt32 size = 512;
MemPtr memPtr;
memPtr = MemPtrNew (size);
```

After executing the given code, the value of `memPtr` is `NULL`. Which statement is correct?

A. A new memory chunk was created.
B. There is insufficient space in the storage heap.
C. There is insufficient space in the dynamic heap.
D. The application must report the error using the statement
`ErrFatalDisplayIf (!memPtr, "Memory corruption: );`

Answer

C. `MemPtrNew` allocates a new nonmovable chunk in the dynamic heap. If there is insufficient space in the dynamic heap, `MemPtrNew` returns 0.

References

For more information on the Memory Manager API functions, see *Palm OS Programmer's API Reference* and the training class Palm OS Programming Essentials. See also Knowledge Base article 1748.

12. Which two statements are true about creating a Graffiti shift indicator at run time? (Choose two.)

A. A form may contain at most three Graffiti shift indicators.
B. `FrmNewGsi` may cause the form to move in memory.
C. Palm OS discourages the use of dynamic user interface elements.
D. There is no API support for adding a Graffiti shift indicator at run time.

Answer

B and **C**. `FrmNewGsi` requires a pointer to a pointer to the form for the dynamic Graffiti shift indicator. The old value of this pointer may not be valid after the API call, so the API call returns a new value.

The preferred method of having UI objects appear as needed is to create the objects in Constructor and set their usable attributes to `false`. Most applications will never need to change any user interface elements at runtime. The built-in applications don't do so, and the Palm user interface guidelines discourage it.

References

For more information on the Form API functions, see *Palm OS Programmer's API Reference*. For information on dynamic user interfaces, see *Palm OS Programmer's Companion*. The training class Palm OS Programming Essentials also covers these topics.

13. Which statement is true about the direct color member in a bitmap family?

    A. It uses a 16-bit color palette.
    B. The red, green and blue values are each specified with an 8-bit integer.
    C. It must be specified if it is a 16-bit direct color member or 24-bit direct color member.
    D. The direct color member must be disabled if there is no direct color support on the handheld.

Answer

B. The `BitmapDirectInfoType` structure uses 8-bit integers to define the number of bits used by the red, green, and blue components in each pixel.

References

For more information about using bitmaps, see *Palm OS Programmer's API Reference* and *Palm OS Programmer's Companion*.

14. Given that controlP contains a pointer to a valid control:

```
char* strP;

strP = MemPtrNew(20);
StrCopy (strP, “new control text”);
CtlSetLabel (controlP, strP);
```

Which statement is true?

    A. The system will automatically free `strP` when the form closes.
    B. The application can safely free `strP` immediately after calling `CtlSetLabel`.
    C. The system will automatically free `strP` the next time `CtlSetLabel` is called.
    D. The application is responsible for freeing `strP` at the end of the control's life.

Answer

D. If you use `CtlSetLabel`, you need to free the string after it is no longer in use (typically after the form containing the control is freed).

References

For more information about user interface elements and the Control API functions, see *Palm OS Programmer's API Reference* and *Palm OS Programmer's Companion*. The training class Palm OS Programming Essentials also covers these topics.

15. Which two statements are true about saved preferences? (Choose two.)

A. The saved preferences are deleted during a soft reset.
B. The saved preferences are stored in the app info block of the application (PRC).
C. The standard Palm OS Launcher will delete the associated saved preferences when a Palm OS application (PRC) is deleted.
D. The saved preferences for a specific creator id are deleted when the associated application is deleted using `DmDeleteDatabase`.
E. The saved preferences for all applications are stored in a single record database (PDB).
F. The saved preferences associated with a Palm OS application (PRC) are beamed with the application.

Answer

C and **E**. Preferences Manager uses one database for "saved" preferences and one database for "unsaved" preferences. Preferences Manager uses the application creator ID to show which application owns the preferences, so when the application is deleted, the associated preferences are also deleted. The preferences are stored in the storage heap, so they survive a soft reset.

References

For more information on the Preferences Manager, see *Palm OS Programmer's API Reference*, *Palm OS Programmer's Companion*, and the training class Palm OS Programming Essentials. This topic is also covered in the online training class Introduction to Palm OS Application Development.

16. Under which three circumstances is the `sysAppLaunchCmdSystemReset` launch code issued? (Choose three.)

A. A soft reset
B. A soft reset while holding down the up arrow key
C. A hard reset after which the user chose to delete all data
D. A hard reset, after which the user chose not to delete all data.

Answer

A, **C**, and **D**. This launch code is sent to applications to allow them to respond to system soft or hard reset. However, for a soft reset while holding down the up arrow key, the sysAppLaunchCmdSystemReset launch code is not sent to applications. This is useful if there is an application on the handheld that crashes upon receiving this launch code (not uncommon) and therefore prevents the system from booting.

References

For more information about launch codes and system resets, see *Palm OS Programmer's API Reference*, *Palm OS Programmer's Companion*, and the training class Palm OS Programming Essentials. See also Knowledge Base articles 1712 and 1713.

17. Upon receiving a `sysAppLaunchCmdFind` launch code, why does it often make sense for your application to pass the length of the matching string in the `appCustom` parameter when calling `FindSaveMatch`?

A. This extra information speeds up the `sysAppLaunchCmdGoto` operation.
B. The global Find facility cannot always determine the length of the matching string.
C. The match string passed in the `sysAppLaunchCmdGoto` parameter block is not null-terminated.
D. The match string length is the length of the entire record, not just the length of the matched portion.

Answer

B. Passing the length of the matching string allows an application to highlight the search string when displaying the resulting record.

References

For more information about launch codes and using global Find, see *Palm OS Programmer's API Reference*, *Palm OS Programmer's Companion*, and the training class Palm OS Programming Essentials.

18. Your application makes use of the Attention Manager. The application receives a `kAttnCommandDrawDetail` command for an attention item that is no longer valid. Which three responses are acceptable? (Choose three.)

A. Call `AttnUpdate`
B. Call `AttnForgetIt`
C. Ignore the command
D. Display an error message to the user
E. Set the `kAttnFlagsNothing` bit in the flags field of the accompanying `AttnFlagsType` structure.

Answer

B, **C**, and **D**. `AttnUpdate` can cause an attention item to redraw, and `kAttnFlagsNothing` disables all attention-getting mechanisms.

References

For a full description of the Attention Manager, see *Palm OS Programmer's Companion*.

19. You want to force periodic updates of your application's Attention Manager items. What should your application do?

A. Set an alarm with `AlmSetAlarm`, and call `AttnUpdate` after receiving an Alarm Manager launch code
B. Set an alarm with `AlmSetProcAlarm`, and call `AttnUpdate` from within the procedure alarm callback function
C. Set an alarm with `AlmSetAlarm`, and call `AttnGetAttention` after receiving an Alarm Manager launch code
D. Set an alarm with `AlmSetProcAlarm`, and call `AttnGetAttention` after receiving an Alarm Manager launch code

Answer

B. `AlmSetAlarm` specifies an application to be called in response to an alarm, but `AlmSetProcAlarm` specifies a procedure to be called. `AttnGetAttention` requests the user's attention, but `AttnUpdate` updates the attention item through the callback procedure.

References

For a full description of the Attention Manager API functions, see *Palm OS Programmer's API Reference*, as well as *Palm OS Programmer's Companion.*

20. Which two statements are true about records in a Palm OS record database and categories in a standard Palm OS category structure? (Choose two.)

    A. Records and categories have a delete bit.
    B. Records have a dirty bit; categories have a renamed bit.
    C. Record unique identifiers and category unique identifiers are each 16-bit values.
    D. A record is secret if either the record or the record category is marked secret.
    E. The category field in the record header contains the category index, not the unique ID.

Answer

B and **E**. Records have a delete bit, but categories don't. The category unique IDs, which are used for synchronization with the desktop database, are 8-bit values. Records can be marked as secret, but categories can't.

References

For more information on records and categories, see *Palm OS Programmer's API Reference* and *Palm OS Programmer's Companion*. The training class Palm OS Programming Essentials also covers these topics. See also Knowledge Base article 1791.

21. Using the Net Library, how many sockets may be open at the same time?

    A. Four
    B. Eight
    C. Four for listening, four for sending
    D. Varies depending on Palm OS version
    E. Varies depending on the amount of memory available

Answer

A. You can open a maximum of four sockets at once in the net library. This requirement is to keep the net library's memory requirements to a minimum.

References

For more information on the Network Communication API functions, see *Palm OS Programmer's Companion, Volume II, Communications*. This topic is also covered in the training class Palm OS Communications Essentials. See also Knowledge Base article 1755.

22. Two applications in Palm OS 4.0 need to register with the Exchange Manager for the same

data type. Which behavior does Palm Recommend?

A. The application that is installed first should receive all objects of the specified type.
B. The application that is installed last should receive all objects of the specified type.
C. The Exchange Manager should be allowed to choose which application will receive all objects of the specified type.
D. Both applications should allow users to choose which application should, by default, receive objects of the specified type.
E. The second application to register with the Exchange Manager will always receive the data. The first application cannot receive the data.

Answer

D. PalmSource, Inc. strongly recommends that you allow users to choose which application is the default. To do so, you could display a panel that shows users the applications that can receive the same type of data as your application, show them which is the default, and allow them to select a different default.

References

For more information on the Exchange Manager API functions, see *Palm OS Programmer's Companion, Volume II, Communications*. This topic is also covered in the training class Palm OS Programming Essentials

23. What should you do to get the date format for a particular handheld?

A. You should use `PrefGetPreference` or `PrefGetPreferences`.
B. The Overlay Manager will return the correct overlay based on the system locale so you can specify the date format for each locale within an overlay for the specific locale.
C. Date formats change based on country so check the system locale with `OmGetCurrentLocale` and then use the Locale Manager to determine the date format for that locale.
D. The date format may be stored differently in the system in future version of the Palm OS. For future compatibility, it is safer to allow the user to set their preferred date format for the application from with in your application.

Answer

A. Palm OS applications should honor the date format set in the system preferences. It is strongly recommended that applications not override the system preferences. (Note that in most cases you should use the function `PrefGetPreference`; `PrefGetPreferences` retrieves the 1.0 version of the system preferences structure.)

References

For information on using preferences, see *Palm OS Programmer's Companion*. For more information on the Preferences API functions, see *Palm OS Programmer's API Reference*. For information on respecting user preferences, see *Palm OS User Interface Guidelines*. and the Knowledge Base articles 1843 and 1726.

24. Which statement best describes the process by which an overlay is validated for a particular database?

    A. The overlay must have the same database name and a type of `'ovly'`.
    B. The base resource database contains a list of database names that specify the appropriate overlay for each locale.
    C. The application must check the current system locale with `OmGetCurrentLocale` and then define the appropriate overlay by giving its unique database name.
    D. In addition to having a database name that matches the overlay naming convention used by the Palm OS, it must have the same creator ID, type of `'ovly'`, and a resource checksum that matches that of the base resource.

    Answer

    D. An overlay database has the same creator as the base database, but its type is `'ovly'`, and a suffix identifying the target locale is appended to its name.

    References

    For information on localized applications, see *Palm OS Programmer's Companion*. For information on creating overlays, see *Palm OS Development Tools Guide*. See also Knowledge Base article 1747.

25. You set the owner ID of a pointer to 0. What effect does this have?

    A. The chunk becomes relocatable.
    B. The chunk cannot be deallocated by your application.
    C. The chunk can no longer be accessed by your application.
    D. The chunk will not be automatically deallocated when your application quits.

    Answer

    D. If the system needs to allocate a chunk that is not disposed of when an application quits, it changes the chunk's owner ID to 0 by calling the system functions MemHandleSetOwner or MemPtrSetOwner.

    References

    For more information on memory use on Palm OS, see *Palm OS Programmer's Companion*. This topic is also covered in the training class Palm OS Programming Essentials.

    See also Knowledge Base article 1076, which gives a code sample.

26. Which two statements are true about handles? (Choose two.)

    A. Handles use less dynamic heap space than pointers.
    B. Handles should be used for data chunks that require frequent resizing.
    C. Handles CANNOT be locked more than once with out unlocking.
    D. Handle values can be safely stored with an application's preferences.
    E. Locked handles are more likely to cause heap fragmentation than pointers.

Answer

D and **E**. A handle stored in a database record or a resource will be reinitialized when the system resets; however, preferences survive a soft reset because they reside in the storage heap.

When an application no longer needs the data chunk, it should unlock the handle immediately to keep heap fragmentation to a minimum.

References

For more information on memory use on Palm OS, see *Palm OS Programmer's Companion*. This topic is also covered in the training courses Introduction to Palm OS Application Development (online) and Palm OS Programming Essentials (instructor-led)

27. Your Palm OS solution does NOT utilize a custom conduit. In which of the following situations will the Backup conduit perform the backup?

    A. The database is of type `'INFO'` and the backup bit is not set
    B. The database is of type `'DATA'` and the backup bit is not set
    C. The database has been modified and the backup bit is set
    D. The database has not been modified and the backup bit is set
    E. The database has been modified and the backup bit is not set

Answer

C. The Backup conduit handles any databases marked for backing up that have not been handled by another conduit.

References

For information about the Backup conduit, see *Introduction to Conduit Development* and the training class Creating Palm OS Conduits on Windows in C/C++. See also Knowledge Base articles 1175, 1103, and 1050.

28. All debugging checks are turned on in the Palm OS Emulator when the following code executes:

```
MemPtr p = MemPtrNew(10);
MemSet(p, 100, 100);
```

What happens on the Palm OS Emulator and what happens on the handheld?

    A. Always works on Emulator; always works on the handheld.
    B. Always crashes on Emulator; always works on the handheld.
    C. Sometimes crashes on Emulator; always works on the handheld.
    D. Always crashes on Emulator; sometimes crashes on the handheld.
    E. Sometimes crashes on Emulator; sometimes crashes on the handheld.

Answer

D. Palm OS Emulator detects memory access exceptions involving access to a memory location that the application is not supposed to access.

References

For more information on the Memory API functions, see *Palm OS Programmer's API Reference*. For information on using Palm OS Emulator, see *Using Palm OS Emulator*. This topic is also covered in the training courses Introduction to Palm OS Application Development (online) and Palm OS Programming Essentials (instructor-led)

29. A global variable in your application gets corrupted during testing. What is the best method for catching the corruption as it occurs?

    A. Attach the Palm Debugger and use the `ss` command.
    B. Attach the debugging console and use the `hf` command.
    C. Turn on all of the Palm OS Emulator debugging preferences.
    D. Use your debugger to set a breakpoint on the global variable.

Answer

A. The Palm Debugger `ss` command allows you to "step-spy" through your application: step until the value of the specified address changes.

References

For information on using Palm Debugger, see *Palm OS Programming Development Tools Guide*. the training course Advanced Palm OS Development and Debugging.

30. Your application runs properly from the debugger on the Palm OS Emulator, but its icon is NOT visible in the application launcher. How is this possible?

    A. The application's hidden bit is cleared.
    B. The application's type ID is set to `'APPL'`.
    C. The application's creator ID is set to `'Palm'`.
    D. The application's type ID matches another application.
    E. The application's creator ID matches another application.

Answer

B. Type IDs are case-sensitive, and the application's type ID should be set to `'appl'`.

References

For information on database type IDs and creator IDs, see *Palm OS Programmer's Companion*. This topic is also covered in the training courses Introduction to Palm OS Application Development (online) and Palm OS Programming Essentials (instructor-led). For users of Metrowerks CodeWarrior for Palm, see the PalmRez post-linker panel settings in the Targeting Palm OS® manual.